

How Does Sandbagging Manifest in Esports FPS Aiming?

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Abstract

Sandbagging—intentionally underperforming to lower expectations—undermines fairness in esports. In esports environments, where nonverbal cues are severely constrained, sandbagging cannot be expressed or observed in the same manner as in physical sports. As a result, how sandbagging manifests as concrete in-game behavior remains insufficiently understood. In this study, we investigate the behavioral characteristics of sandbagging through an analysis of operation logs collected from a controlled first-person shooter aiming task. Ten participants performed the task under both high-effort and sandbagging conditions, during which detailed metrics such as shot counts and hit counts were recorded. Our analysis reveals that esports players may adopt at least two distinct sandbagging strategies. One strategy involves consistently suppressing the number of shots while largely avoiding intentional misses. The other strategy is characterized by behavioral switching across trials, alternating between suppressing shot counts and deliberately increasing missed shots without reducing firing frequency. This work provides foundational empirical insights into sandbagging behavior and highlights key directions for the development of automated sandbagging detection methods in esports.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Sandbagging, Esports

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1 Introduction

Sandbagging is a self-presentational strategy in which individuals deliberately induce others to underestimate their performance expectations by falsely claiming low ability or engaging in feigned demonstrations of inability [5, 8]. This behavior is more likely to be exhibited by individuals with high public self-consciousness, particularly in situations where others hold high expectations for their success. Moreover, prior work suggests that the tendency to engage in sandbagging is amplified when others lack knowledge of the individual's past performance or achievements [6]. Concretely, sandbagging has been observed in different forms across contexts. In educational settings, it often appears as students publicly claiming, prior to an examination, that they have not studied at all, thereby lowering others' expectations of their performance [5]. In physical sports contexts, sandbagging may be enacted through nonverbal behaviors such as adopting a lack-of-confidence facial expression, pretending to be fatigued, or intentionally displaying awkward or unrefined technique [6]. Thus, unlike self-handicapping, which involves intentionally reducing one's own preparation or training in order to preserve self-acceptance in the event of failure [2, 15], sandbagging can be characterized as a more sophisticated, outward-facing strategy that seeks to increase one's advantage in competitive situations by strategically manipulating others' expectations.

In esports, a form of competitive sport mediated by video games, the possibility of sandbagging has also been discussed in qualitative analyses of player behavior in competitive online games [7], as well as in community discourse and informal reports [10, 16]. However, because esports are typically conducted in online environments with limited communication channels, the availability of nonverbal cues expressed by players is severely constrained. As a result, sandbagging in esports cannot be observed in the same forms as in physical sports, such as visibly displaying fatigue or confidence-related facial expressions. In other words, to the best of our knowledge, while there have been cases of detecting affiliation [3] and acts of insulting other players [1, 9, 14], how sandbagging manifests as concrete operations or behaviors on a digital game screen in esports remains insufficiently explored, leaving this issue as a largely uncharted research gap. This gap is not only of academic interest but also raises important concerns regarding the fairness of esports competitions. In esports tournaments that involve monetary rewards or social prestige, match seeding and handicap systems are often designed based on estimations of participants' skill levels. Sandbagging has the potential to not only



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undermine these mechanisms but also fundamentally damage the ethos of the competition [4, 12], as it allows players to obtain unfair advantages by deceiving systems intended to ensure competitive balance. From the perspective of maintaining competitive integrity, sandbagging should therefore not be regarded as a merely permissible strategic choice, but rather as a behavior that warrants active detection and deterrence.

Motivated by this perspective, this study investigates how sandbagging manifests behaviorally in esports and discusses the feasibility of its automatic detection. Specifically, we focus on aiming behavior, which constitutes a core and performance-critical operation in first-person shooter (FPS) games, one of the most representative genres in esports. We analyze how players' behaviors change when they engage in sandbagging during aiming tasks. The findings and discussion of this paper elucidate behavioral manifestations of sandbagging in FPS aiming tasks and contribute foundational insights toward the automatic detection of sandbagging in this domain.

2 Data collection

This section details the methodology for analyzing behavioral differences between the high-effort and sandbagging conditions in esports. Hereafter, the high-effort condition is denoted as **HE**, and the sandbagging condition as **SB**.

In FPS games, aiming is a core mechanic that directly determines player performance. Accordingly, we focused on collecting both player performance and input during aiming tasks. To minimize external confounding factors, we developed a custom experimental game in Unity that isolated aiming operations, as shown in the left side of Fig. 1. The game excluded complex elements such as player movement and enemy attacks. Instead, players were required to continuously shoot at targets that appeared sequentially at varying positions for one minute. During gameplay, the number of shots fired, the numbers of hits, and mouse movement speed were automatically recorded. To ensure consistency, all participants used identical hardware, including a PC, a 1920×1080 monitor, and a mouse set to 800 DPI.

Ten male undergraduate and graduate students participated in the data collection (mean age = 22.1 years, SD = 1.45). Hereafter, participants are referred to as **P1–P10**. Regarding FPS experience, P1–P6 had not played any FPS games within the past month, P7 and P8 played no more than twice per week, and P9 and P10 played three or more times per week. After a three-minute practice session, participants completed a total of six trials—three under the HE condition and three under the SB condition—in a randomized order. Under the HE condition, participants were instructed to play as quickly and accurately as possible. In contrast, to prevent extreme underperformance that would deviate from realistic sandbagging behavior, participants in the SB condition were instructed to underperform in a manner that would not be noticeable to others when viewing the recorded gameplay.

3 Result

3.1 Participant-averaged Performance Analysis

Number of shots. Fig. 1 (a) presents the average number of shots across three trials for each participant under both conditions. The mean ($\pm$ SD) shot counts for HE and SB were 89.9 ± 3.03 and $73.1 \pm$

12.8 , respectively. A paired t-test revealed a significant difference between the two groups ($p < .05$). Across both conditions, no strong correlation was observed between FPS play frequency (P1–6: less than once a month; P7–8: twice a week or less; P9–10: three times a week or more) and the number of shots. In the SB condition, the fact that all participants except for P10 reduced their shot counts is consistent with the intended goal of sandbagging. The larger SD of shot counts in the SB condition compared to the HE condition may reflect the diversity of sandbagging strategies adopted by the participants.

Number of hits. Fig. 1 (b) illustrates the average number of hits per participant across three trials. The mean ($\pm$ SD) hit counts were 86.9 ± 6.96 for HE and 62.4 ± 9.10 for SB, with a paired t-test showing a significant difference between the conditions ($p < .05$). Across both conditions, no clear relationship was observed between FPS play frequency and the number of hits. In the SB condition, the reduction in hit counts by all participants is inherently consistent with the nature of the sandbagging task. Notably, the SD of hit counts under the SB condition was smaller than that of shot counts. This disparity likely arises because, while shot counts can be easily adjusted depending on individual sandbagging strategies, participants had no choice but to minimize their hit counts to achieve an intentional underperformance.

Accuracy. Fig. 1 (c) displays the average accuracy, computed as the ratio of hits to shots and averaged across three trials. For HE and SB, the mean accuracy ($\pm$ SD) was 97.0 ± 1.76 and 88.2 ± 7.63 , respectively. A paired t-test confirmed that this difference was statistically significant ($p < .05$). Consistent with the aforementioned metrics, no strong correlation was found between FPS play frequency and accuracy across both conditions. It can be observed that nearly all participants, with the notable exception of P8, tended to lower their accuracy as a means of achieving sandbagging. The specific behavior of P8 will be further analyzed in Section 4.

Mouse movement speed. Fig. 1 (d) shows the average mouse movement speed for each participant. The average speed was 540.0 ± 80.7 pixel/sec in HE and 443.6 ± 135.1 pixel/sec in SB. The difference between the two groups was found to be significant by a paired t-test ($p < .05$). As with the previous metrics, FPS play frequency did not exert a significant influence on this measure. Almost all participants reduced their movement speed in the SB condition, with the sole exception of P2. Notably, P2 exhibited a distinct behavior by substantially increasing the mouse movement speed under the SB condition, a marked departure from the overall trend.

3.2 Trial-wise Behavioral Analysis Across Participants

Plotting the number of shots and hits on a trial-wise basis, as shown in Fig. 2, clarifies the consistency of each player's behavior. In the HE condition, most trials are concentrated near the line $y = x$ (i.e., accuracy ≈ 1.0). However, the degree of proximity between the three trials varies among individuals, regardless of their FPS play frequency. For example, despite both playing twice a week or less, P7's three data points are widely dispersed, whereas those of P8 are relatively clustered. Given that the experimental game involves negligible elements of luck, this disparity may reflect the influence

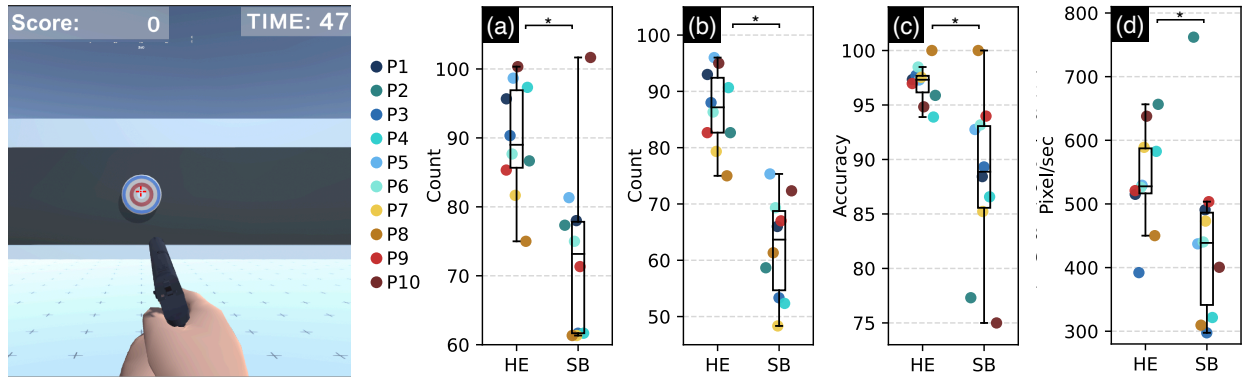


Figure 1: Left: In-game screen of the shooting task. Right: Comparison of metrics between HE and SB: (a) number of shots, (b) number of hits, (c) accuracy, and (d) mouse movement speed. Each data point represents the mean of three trials per participant ($N = 10$).

of skill acquisition or adaptation during the session. In contrast, the SB condition reveals clear evidence of intentional behavioral shifts. Specifically, the trials for participants P1–4, P7, and P8 are plotted significantly further from the $y = x$ line, indicating a marked decrease in accuracy.

4 Discussion

4.1 Strategic Variants of Sandbagging Behavior

Since the experimental results revealed that sandbagging strategies are not monolithic, we re-plotted the data from Fig. 2 on a per-participant basis in Fig. 3 to facilitate a more granular discussion. This visualization allows us to categorize the participants into the following two distinct groups:

- Group 1 (P5, P6, P8, P9): *Consistent shot-suppression type*. This group consistently reduced their shot counts while maintaining relatively high accuracy.
- Group 2 (P1–4, P7, P10): *Strategic hybrid type*. This group employed a mixture of strategies, alternating between reducing shot counts without accuracy loss and significantly degrading their accuracy across trials.

In Group 1, the tendency to avoid a drastic decline in accuracy—the most explicit performance metric in our experimental task—aligns with existing findings on sandbagging [5, 13]. These studies suggest that sandbagging is often executed as a strategic form of self-presentation, where individuals prioritize appearing “plausibly weak” to lower expectations rather than appearing completely incompetent through blatant or unrealistic failure. Indeed, the post-experimental questionnaires regarding participants’ sandbagging approaches corroborated these observations. Participants in this group consistently reported that they adopted an approach to “shoot slowly” (P5, P6, P8, P9) in most trials, whereas there were virtually no reports of intentionally increasing missed shots as a primary tactic. More strictly speaking, this group could be further subdivided into two sub-types based on their specific tactics. Participants P5, P6, and P9, who mentioned that they “intentionally shot at locations other than the target,” could be categorized as a type that incorporates explicit misses, albeit to a lesser extent than

Group 2. In contrast, P8, who adopted the approach of “*aiming at the periphery of the target rather than its center*,” represents a type that avoids explicit misses entirely, opting instead for a subtle reduction in shot quality while technically maintaining accuracy.

For Group 2, the motivation behind trials where accuracy remained high likely aligns with that of Group 1—the desire to maintain a facade of competence while suppressing overall output. However, in a task as simple and deterministic as our experimental aiming game, the decision to alternate between maintaining and significantly degrading accuracy is counterintuitive. Such behavioral inconsistency reveals a lack of sustained engagement, which directly contradicts the fundamental nature of sandbagging: the strategic effort to remain undetected [5, 8]. Furthermore, this volatility is inconsistent with the experimental instruction that participants should avoid letting viewers perceive their lack of effort. We interpret these behaviors as evidence that participants in Group 2 were engaged in a process of “trial and error” to determine the most effective way to sandbag. The post-experimental questionnaires support this interpretation, revealing that they deliberately altered their tactical approach for each trial. For instance, P1 systematically switched between three different methods: “*intentionally acting panicked*,” “*operating with excessive momentum*,” and “*occasionally missing on purpose*.” Similarly, P2 assigned a distinct approach to each trial, such as “*changing the mouse grip*,” “*returning the reticle to a fixed position after every shot*,” and “*following non-optimal trajectories to the target*.” Other participants in Group 2 likewise reported similar shifts in their tactical execution across trials.

While our study identified these two distinct groups, the universality of this tendency warrants further investigation. In this experiment, the novelty of the task likely prompted Group 2 participants to engage in the observed trial-and-error behavior. As players gain familiarity with the task, they may discover their own optimal sandbagging methods, potentially ceasing the frequent shifting of tactical approaches between trials. While some of these participants might eventually converge toward the consistent behaviors seen in Group 1, our current findings cannot exclude the possibility that increased task mastery could give rise to entirely new types of sandbagging strategies.

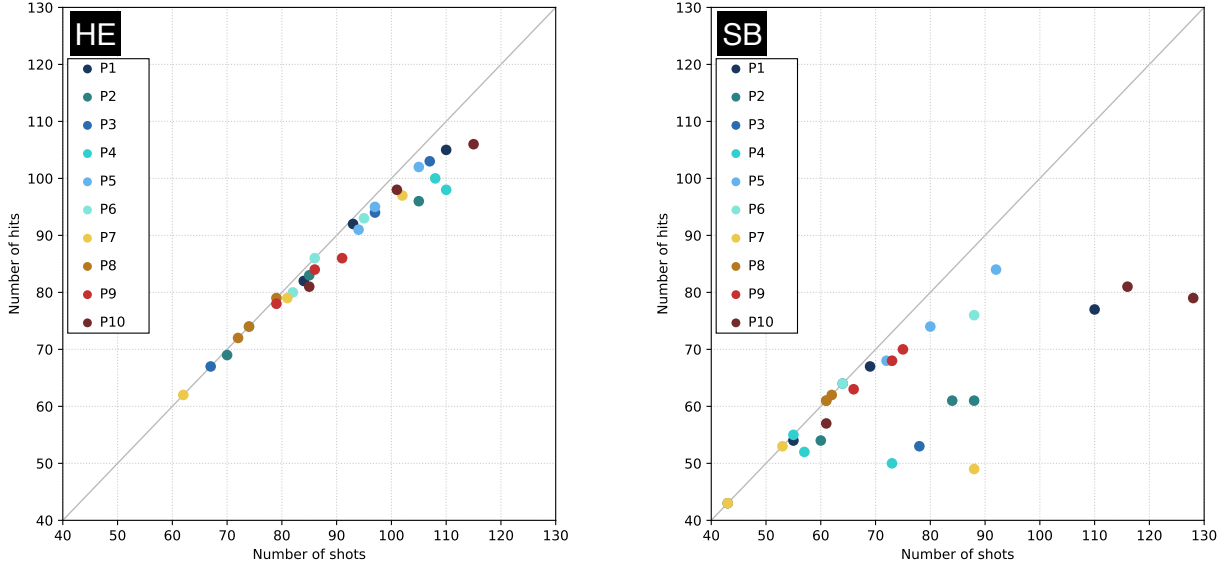


Figure 2: Scatter plot of the number of shots and hits for both HE and SB (10 participants $\times$ 3 trials each).

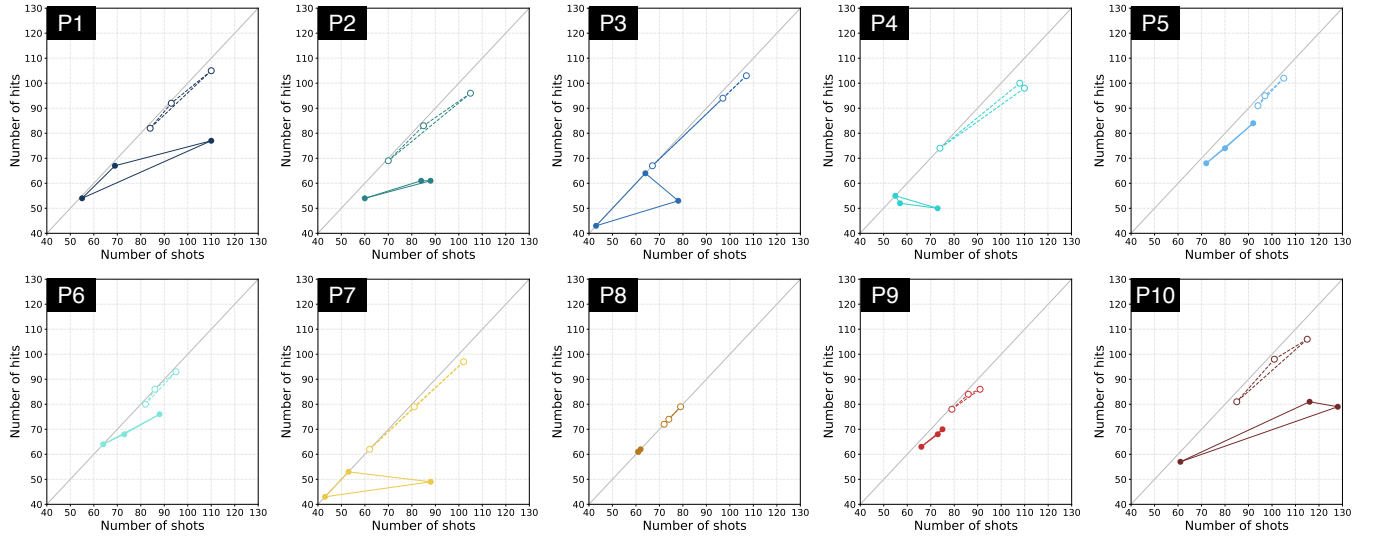


Figure 3: Individual performance across HE and SB conditions. Scatter plots show the number of shots versus hits for 10 participants (3 trials each). Open and filled symbols denote HE and SB conditions, respectively. Lines connect trials within the same condition, illustrating the behavioral shifts and variance between the two conditions.

4.2 Implications for Automated Detection of Sandbagging

Existing approaches for detecting toxic behaviors in esports often employ a single machine learning model that integrates multiple behavioral features into a unified classifier [1, 11]. However, as discussed above, sandbagging does not manifest uniformly. Our findings indicate that at least two distinct strategies can be observed, suggesting that sandbagging behavior spans a broader behavioral space than is easily captured by a single model. This observation

implies that sandbagging detection may benefit from an ensemble-based approach, in which multiple models independently assess the presence of different strategic patterns, rather than relying on a single integrated classifier.

For sandbagging behaviors exemplified by Group 1, where players consistently reduce performance while largely preserving their play style, a natural starting point is to evaluate whether the observed performance deviates from the player's long-term historical performance distribution. However, because temporary declines can also result from momentary mistakes or fluctuations in condition, false positives remain a concern. Incorporating additional

behavioral features, such as mouse movement speed or trajectory characteristics, may help distinguish intentional underperformance from incidental variation, indicating an important direction for future work. In contrast, sandbagging behaviors characteristic of Group 2 involve pronounced short-term changes in play style or performance. For such cases, detection may be more effectively grounded in analyzing the short-term stability of recent gameplay. Abrupt inconsistencies in performance metrics or operational characteristics across consecutive sessions can serve as important cues for intentional behavioral shifts, enabling more effective identification of episodic or exploratory sandbagging.

Taken together, these considerations highlight that detecting sandbagging may require complementary strategies that explicitly account for strategic diversity. This perspective motivates the exploration of multi-model or ensemble-based detection frameworks tailored to the heterogeneous nature of sandbagging behavior.

5 Conclusion

In this study, we analyzed operation logs collected from an FPS aiming task and demonstrated that esports players may adopt at least two distinct sandbagging strategies. One strategy is characterized by a consistent suppression of the number of shots while largely avoiding intentional missed shots. The other strategy involves switching between different behaviors across trials, such as suppressing the number of shots without increasing misses, or alternatively maintaining the shot count while deliberately increasing missed shots. This heterogeneity in behavioral strategies suggests that sandbagging cannot be adequately captured by a single detection model, and instead points to the necessity of sandbagging detection approaches that employ strategy-specific models tailored to different behavioral patterns.

This work represents an exploratory phase of research and is subject to several limitations. First, the target task was restricted to isolated aiming operations and did not incorporate elements inherent to real FPS gameplay, such as player movement, combat with opponents, or situational decision-making. Second, we did not objectively examine whether the observed sandbagging behaviors were truly imperceptible to third-party observers. Future work will address these limitations by collecting larger-scale datasets that incorporate richer gameplay factors and by evaluating sandbagging from both behavioral logs and third-party perspectives. Based on such data, we aim to develop machine learning models capable of detecting sandbagging behaviors in realistic esports environments.

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